STUDENT MANAGEMENT SYSTEM

```
public class Student {
private String name;
private int id;
private int age;
private String department;
Student()
{
name ="";
id=1;
age=1;
department="";
}
Student(String n,int i,int a,String d)
{
name = n;
id = i;
age = a;
department = d;
}
public String getName()
{
return name;
}
public void setName(String n)
{
name = n;
}
public int getID()
{
return id;
}
public void setID(int i)
{
id = i;
}
public int getAge()
{
return age;
}
public void setAge(int a)
{
age = a;
}
public String getDepartment()
{
return department;
}
public void setDepartment(String d)
{
department = d;
}
```

```
public String toString()
{
return "ID: " + id + ", Name: " + name +
", Age: " + age + ", Department: " + department;
}
}

import java.util.ArrayList;
import java.util.Scanner;
public class StudentManagementSystem {
ArrayList<Student> students = new ArrayList<Student>();
public void addStudent(Student s)
{
students.add(s);
System.out.println("Student added successfully");
}
public void displayStudent()
{
if(students.isEmpty())
{
System.out.println("Student not found");
}
else {
System.out.println("The Students are: ");
for(int i=0;i<students.size();i++) {
System.out.print(students.get(i));
}
}
}
public void searchStudent(int id) {
boolean found = false;
for(int i=0;i<students.size();i++){
if(students.get(i).getID() == id) {
System.out.println("Student found: "+students.get(i));
found = true;
break;
}
}
if(!found) {
System.out.println("Student not found");
}
}
public void deleteStudent(int id) {
boolean found = false;
for(int i=0;i<students.size();i++) {
if(students.get(i).getID() == id) {
System.out.println("Student found to be deleted: "+students.get(i));
students.remove(i);
found = true;
break;
}
}
if(!found)
{
System.out.println("Student not found");
}
}
public void updateStudent(int id, String name, int age, String department)
{
```

```
boolean found = false;
for(int i = 0; i < students.size(); i++) {
if(students.get(i).getID() == id) {
students.get(i).setName(name);
students.get(i).setAge(age);
students.get(i).setDepartment(department);
System.out.println("Student updated successfully: " + students.get(i));
found = true;
break;
}
}
if(!found) {
System.out.println("Student not found.");
}
}
public static void main(String[] args) {
StudentManagementSystem system = new StudentManagementSystem();
Scanner sc = new Scanner(System.in);
int choice;

while(true) {
System.out.println("\n=== Student Management System Menu ===");
System.out.println("1. Add Student");
System.out.println("2. Display All Students");
System.out.println("3. Search Student by ID");
System.out.println("4. Delete Student by ID");
System.out.println("5. Update Student");
System.out.println("6. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch(choice) {
case 1:
sc.nextLine(); // consume newline
System.out.print("Enter name: ");
String name = sc.nextLine();
System.out.print("Enter ID: ");
int id = sc.nextInt();
System.out.print("Enter age: ");
int age = sc.nextInt();
sc.nextLine(); // consume newline
System.out.print("Enter department: ");
String dept = sc.nextLine();
Student s = new Student(name, id, age, dept);
system.addStudent(s);
break;
case 2:
system.displayStudent();
break;
case 3:
System.out.print("Enter ID to search: ");
int searchId = sc.nextInt();
system.searchStudent(searchId);
break;
case 4:
System.out.print("Enter ID to delete: ");
int deleteId = sc.nextInt();
system.deleteStudent(deleteId);
break;
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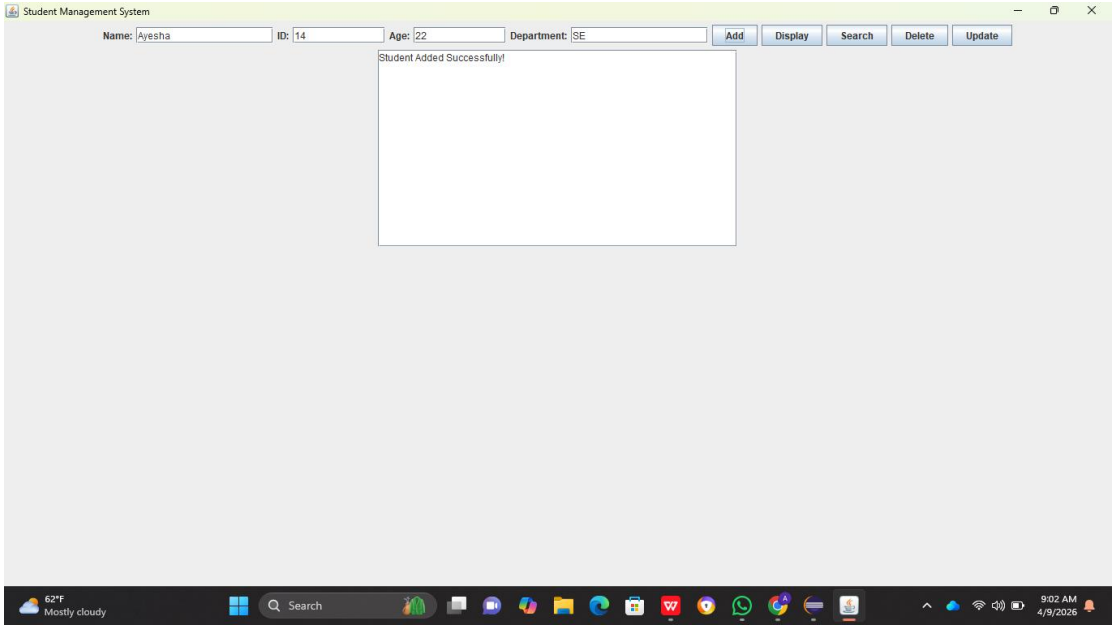
```java
case 5:
System.out.print("Enter ID to update: ");
int updateId = sc.nextInt();
sc.nextLine(); // consume newline
System.out.print("Enter new name: ");
String newName = sc.nextLine();
System.out.print("Enter new age: ");
int newAge = sc.nextInt();
sc.nextLine(); // consume newline
System.out.print("Enter new department: ");
String newDept = sc.nextLine();
system.updateStudent(updateId, newName, newAge, newDept);
break;
case 6:
System.out.println("Exiting the program. Goodbye!");
sc.close();
System.exit(0);
default:
System.out.println("Invalid choice! Try again.");
}
}
}
}
```

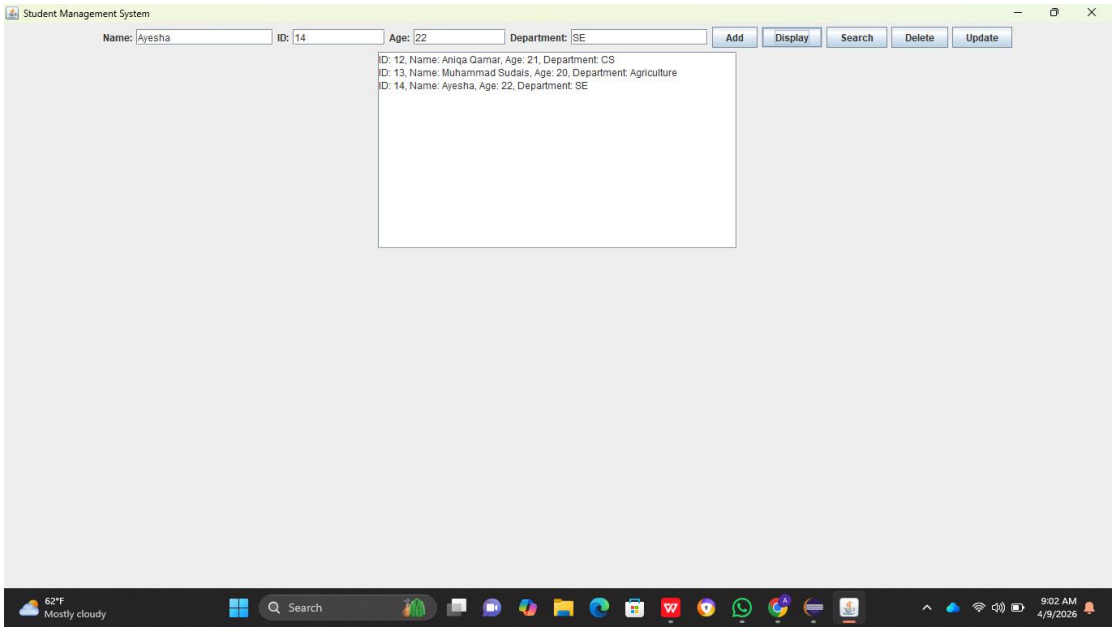
```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.util.ArrayList;
public class StudentGUI extends JFrame{
StudentManagementSystem system = new StudentManagementSystem();

JTextField nameField, idField, ageField, deptField;
JTextArea displayArea;

public StudentGUI() {
setTitle("Student Management System");
setSize(500, 500);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new FlowLayout());

// Input Fields
add(new JLabel("Name:"));
nameField = new JTextField(15);
add(nameField);

add(new JLabel("ID:"));
idField = new JTextField(10);
add(idField);

add(new JLabel("Age:"));
ageField = new JTextField(10);
add(ageField);

add(new JLabel("Department:"));
deptField = new JTextField(15);
add(deptField);

// Buttons
```

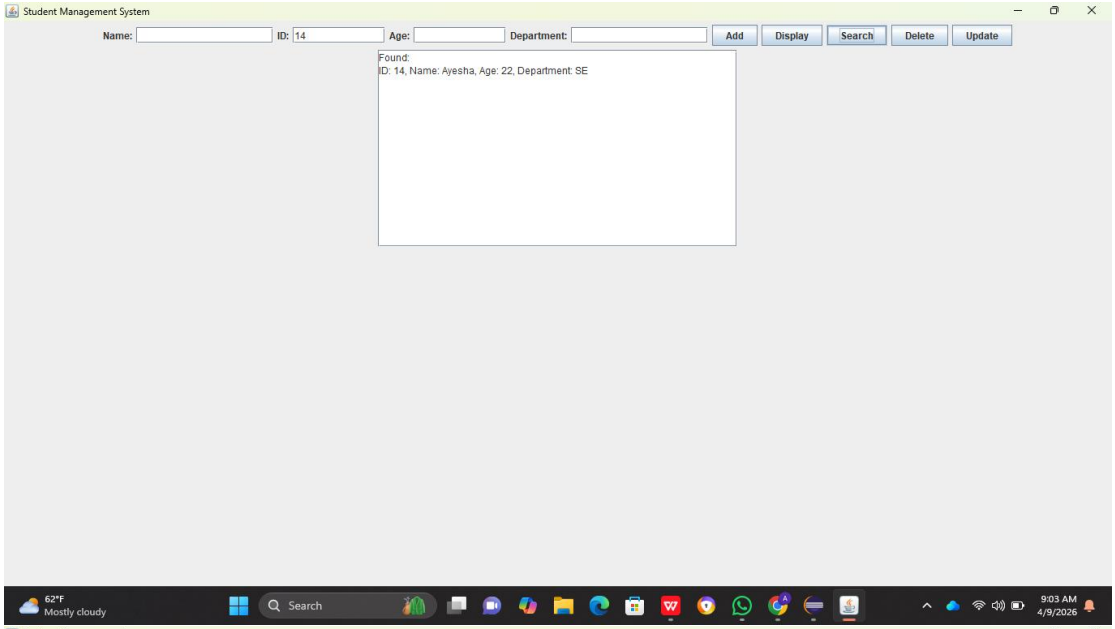
```java
JButton addBtn = new JButton("Add");
JButton displayBtn = new JButton("Display");
JButton searchBtn = new JButton("Search");
JButton deleteBtn = new JButton("Delete");
JButton updateBtn = new JButton("Update");

add(addBtn);
add(displayBtn);
add(searchBtn);
add(deleteBtn);
add(updateBtn);

// Display Area
displayArea = new JTextArea(15, 40);
displayArea.setEditable(false);
add(new JScrollPane(displayArea));

//   BUTTON ACTIONS

// ADD
addBtn.addActionListener(new ActionListener() {
public void actionPerformed(ActionEvent e) {
String name = nameField.getText();
int id = Integer.parseInt(idField.getText());
int age = Integer.parseInt(ageField.getText());
String dept = deptField.getText();

Student s = new Student(name, id, age, dept);
system.addStudent(s);

displayArea.setText("Student Added Successfully!\n");
}
});

// DISPLAY
displayBtn.addActionListener(new ActionListener() {
public void actionPerformed(ActionEvent e) {
displayArea.setText("");
if(system.students.isEmpty()) {
displayArea.setText("No students found.");
} else {
for(int i = 0; i < system.students.size(); i++) {
displayArea.append(system.students.get(i) + "\n");
}
}
}
});

// SEARCH
searchBtn.addActionListener(new ActionListener() {
public void actionPerformed(ActionEvent e) {
int id = Integer.parseInt(idField.getText());
boolean found = false;

for(int i = 0; i < system.students.size(); i++) {
if(system.students.get(i).getID() == id) {
displayArea.setText("Found:\n" + system.students.get(i));
```

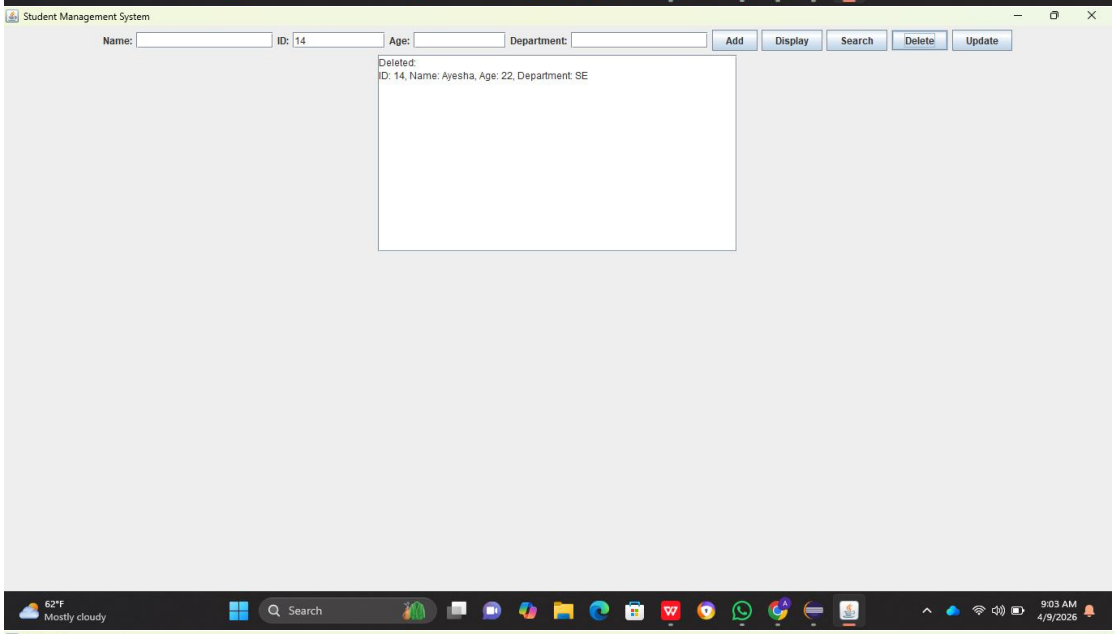
```
found = true;
break;
}
}

if(!found) {
displayArea.setText("Student not found.");
}
}
});

// DELETE
deleteBtn.addActionListener(new ActionListener() {
public void actionPerformed(ActionEvent e) {
int id = Integer.parseInt(idField.getText());
boolean found = false;

for(int i = 0; i < system.students.size(); i++) {
if(system.students.get(i).getID() == id) {
displayArea.setText("Deleted:\n" + system.students.get(i));
system.students.remove(i);
found = true;
break;
}
}

if(!found) {
displayArea.setText("Student not found.");
}
}
});

// UPDATE
updateBtn.addActionListener(new ActionListener() {
public void actionPerformed(ActionEvent e) {
int id = Integer.parseInt(idField.getText());
String name = nameField.getText();
int age = Integer.parseInt(ageField.getText());
String dept = deptField.getText();

boolean found = false;

for(int i = 0; i < system.students.size(); i++) {
if(system.students.get(i).getID() == id) {
system.students.get(i).setName(name);
system.students.get(i).setAge(age);
system.students.get(i).setDepartment(dept);

displayArea.setText("Updated:\n" + system.students.get(i));
found = true;
break;
}
}

if(!found) {
displayArea.setText("Student not found.");
}
```

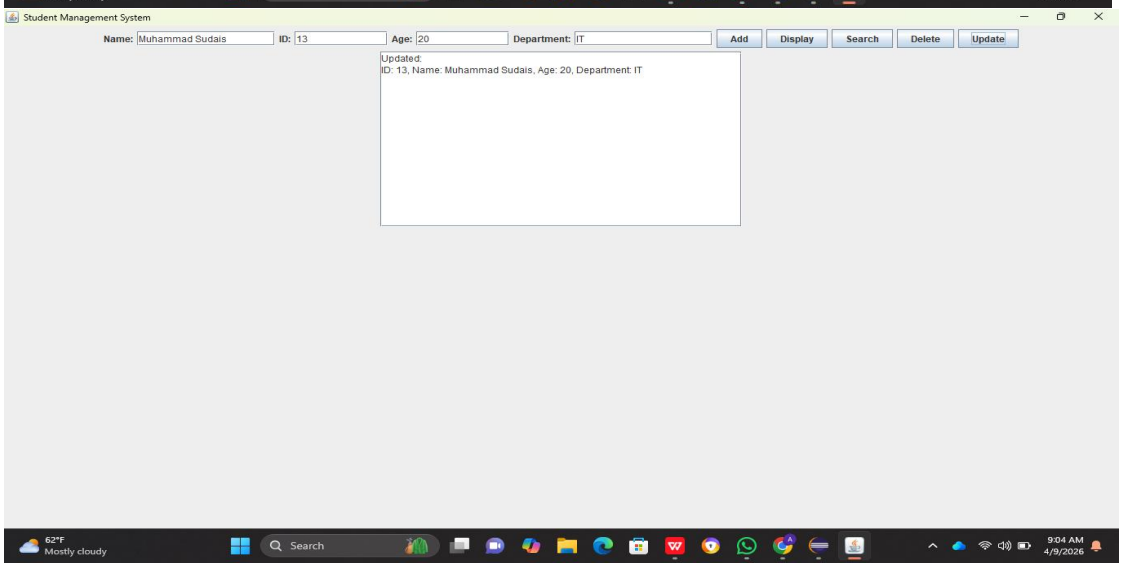
```
}
});
}

public static void main(String[] args) {
new StudentGUI().setVisible(true);
}
}
```






Student Management System
Name:
ID: 14
Age:
Department:
Add
Display
Search
Delete
Update
Found:
ID: 14, Name: Ayesha, Age: 22, Department: SE
62°F
Mostly cloudy
Search
9:03 AM
4/9/2026


Student Management System
Name:
ID: 14
Age:
Department:
Add
Display
Search
Delete
Update
Deleted:
ID: 14, Name: Ayesha, Age: 22, Department: SE
62°F
Mostly cloudy
Search
9:03 AM
4/9/2026


Student Management System
Name: Muhammad Sudais
ID: 13
Age: 20
Department: IT
Add
Display
Search
Delete
Update
Updated:
ID: 13, Name: Muhammad Sudais, Age: 20, Department: IT
62°F
Mostly cloudy
Search
9:04 AM
4/9/2026